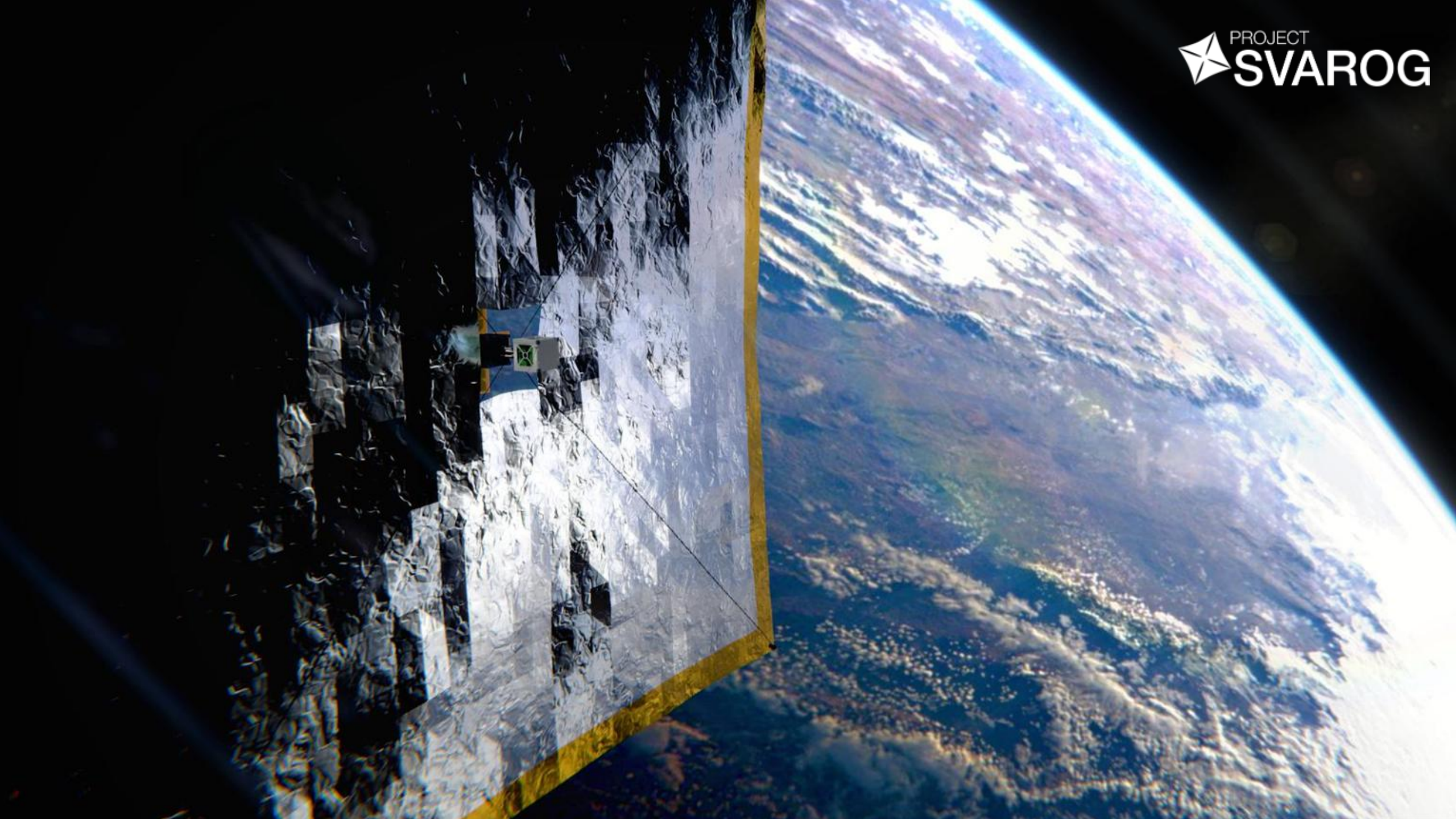
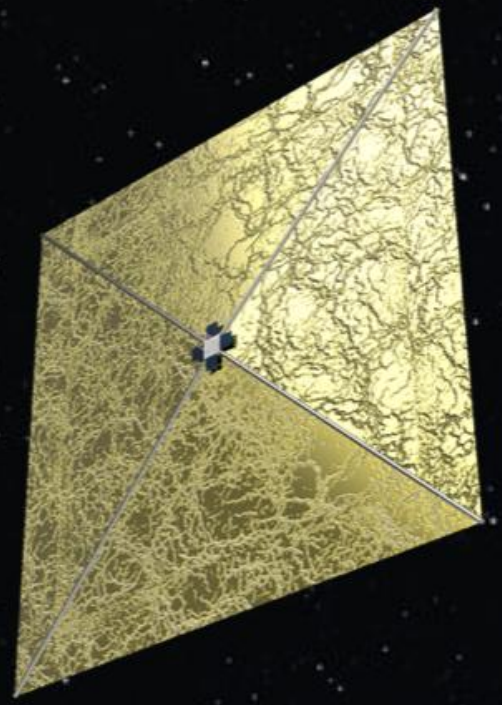
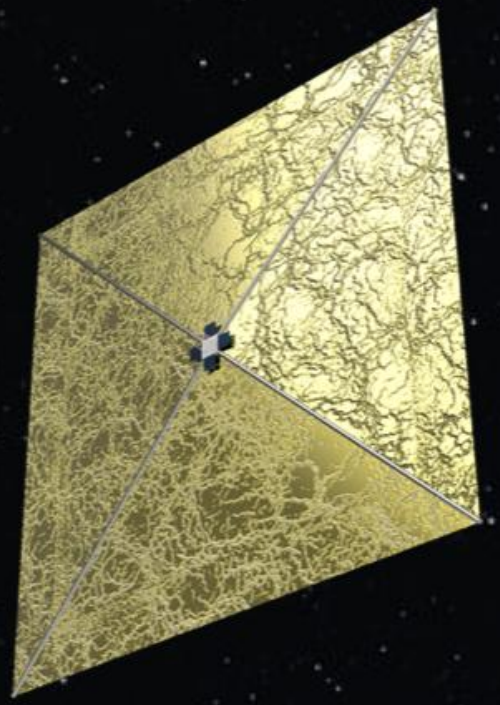

PROJECT
SVAROG

Welcome to



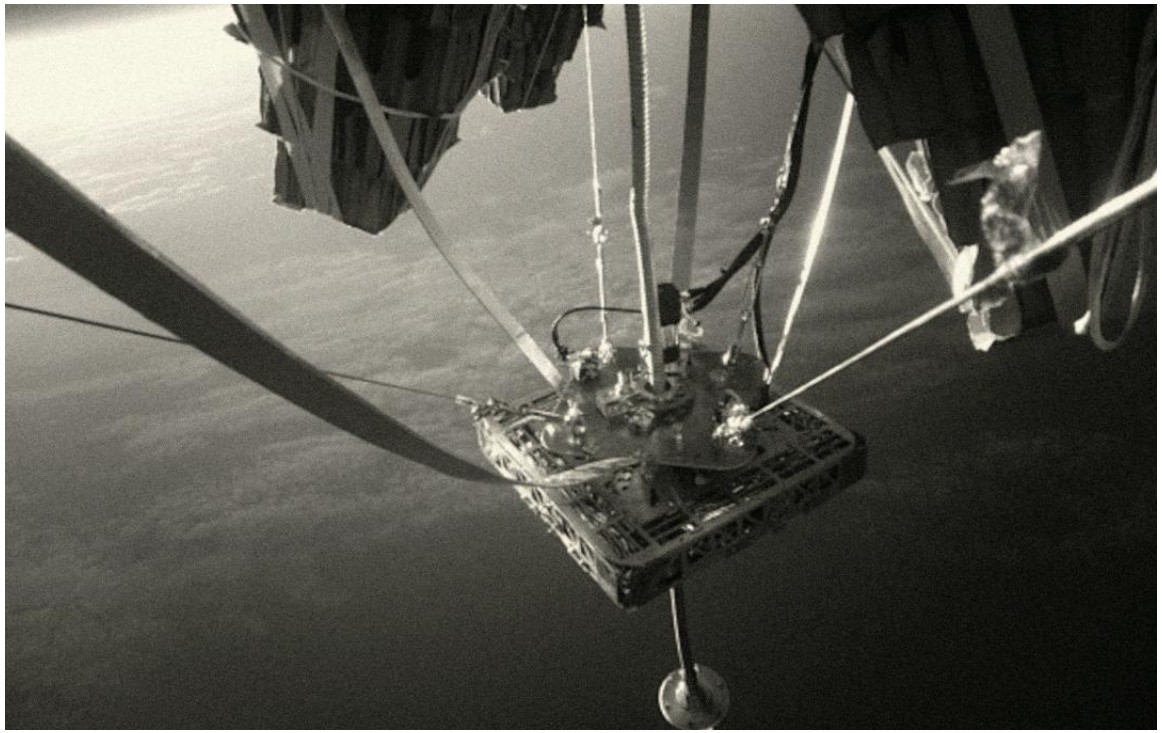
Project Svarog

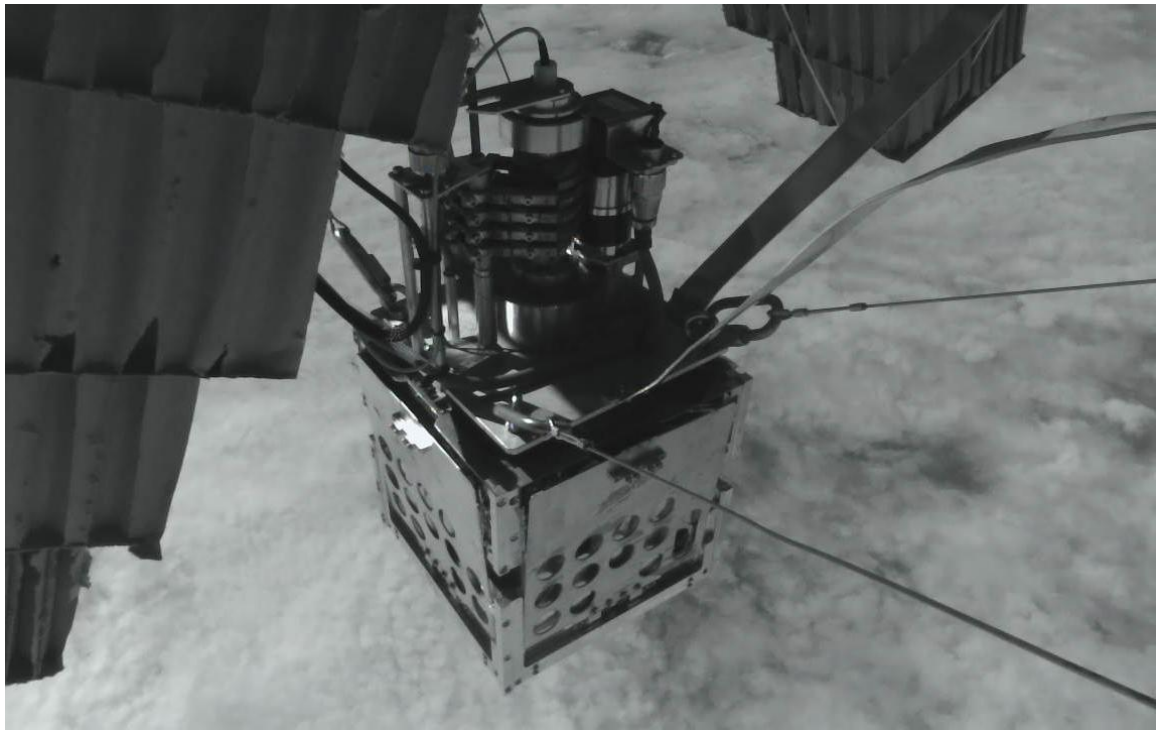
Our goal



is to launch a solar-sail powered spacecraft that will one day leave the solar system

What have we done so far?





BEXUS-34

October 2024

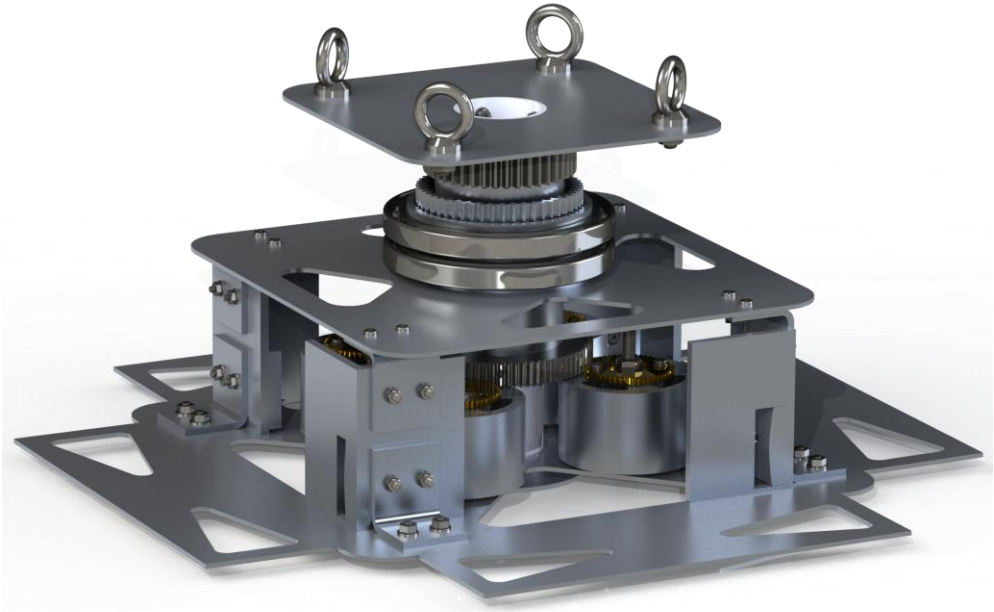
BEXUS-36

August 2026

BEXUS-34

Launched 7th October 2024 from Esrange Space Center, Sweden

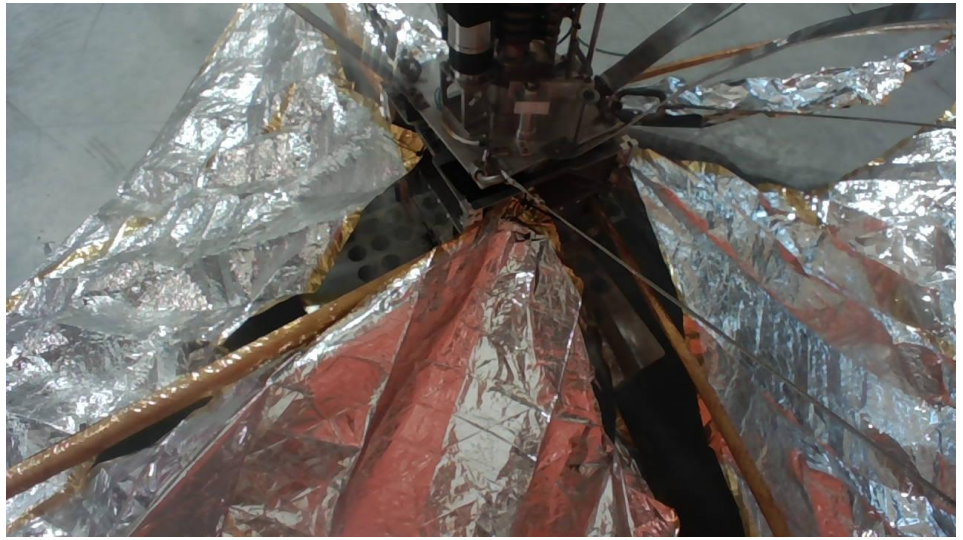




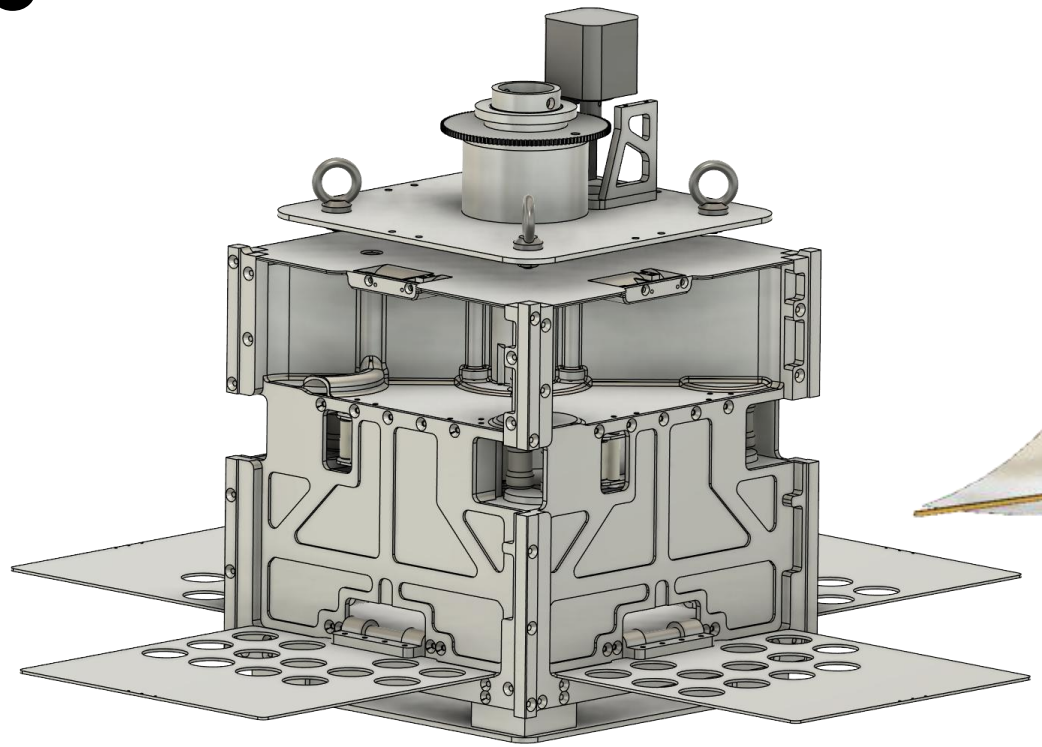


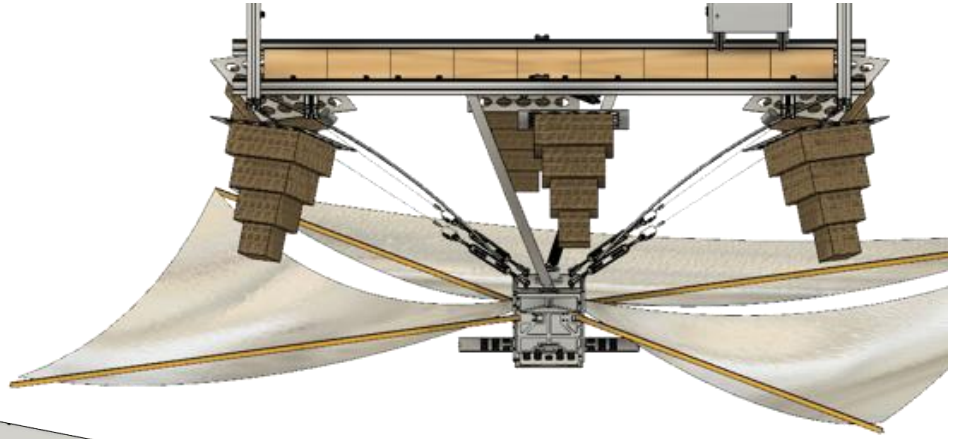
BEXUS-36

Launched ~~8th October 2025~~ 12th August 2026 from Esrange Space Center, Sweden







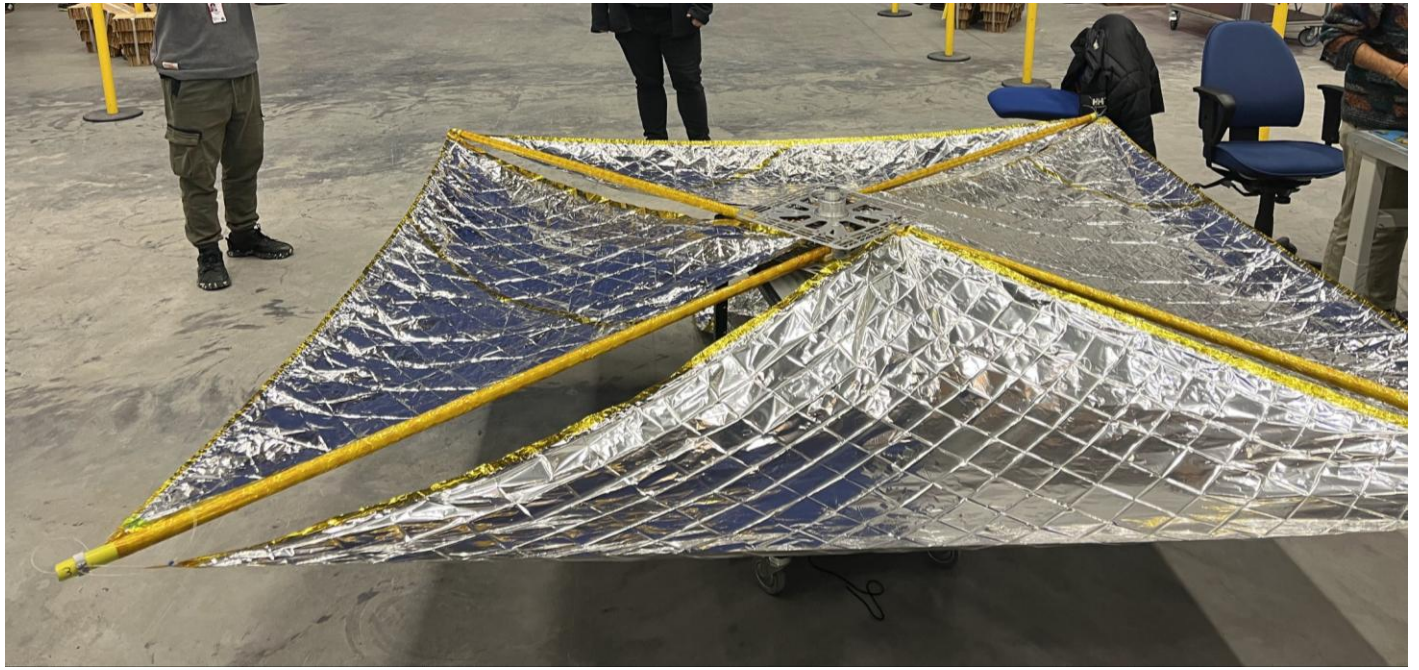


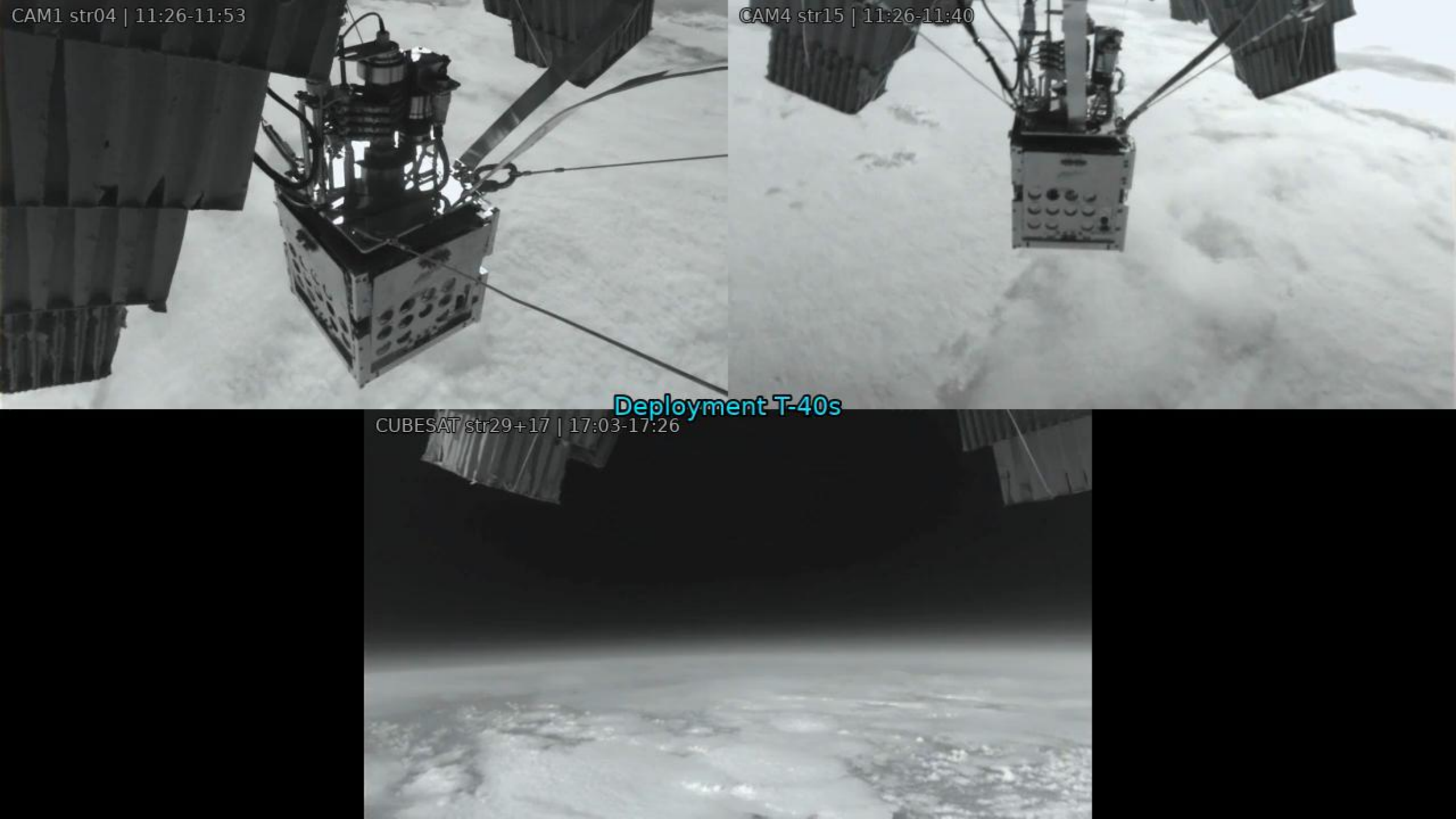

PROJECT SVAROG





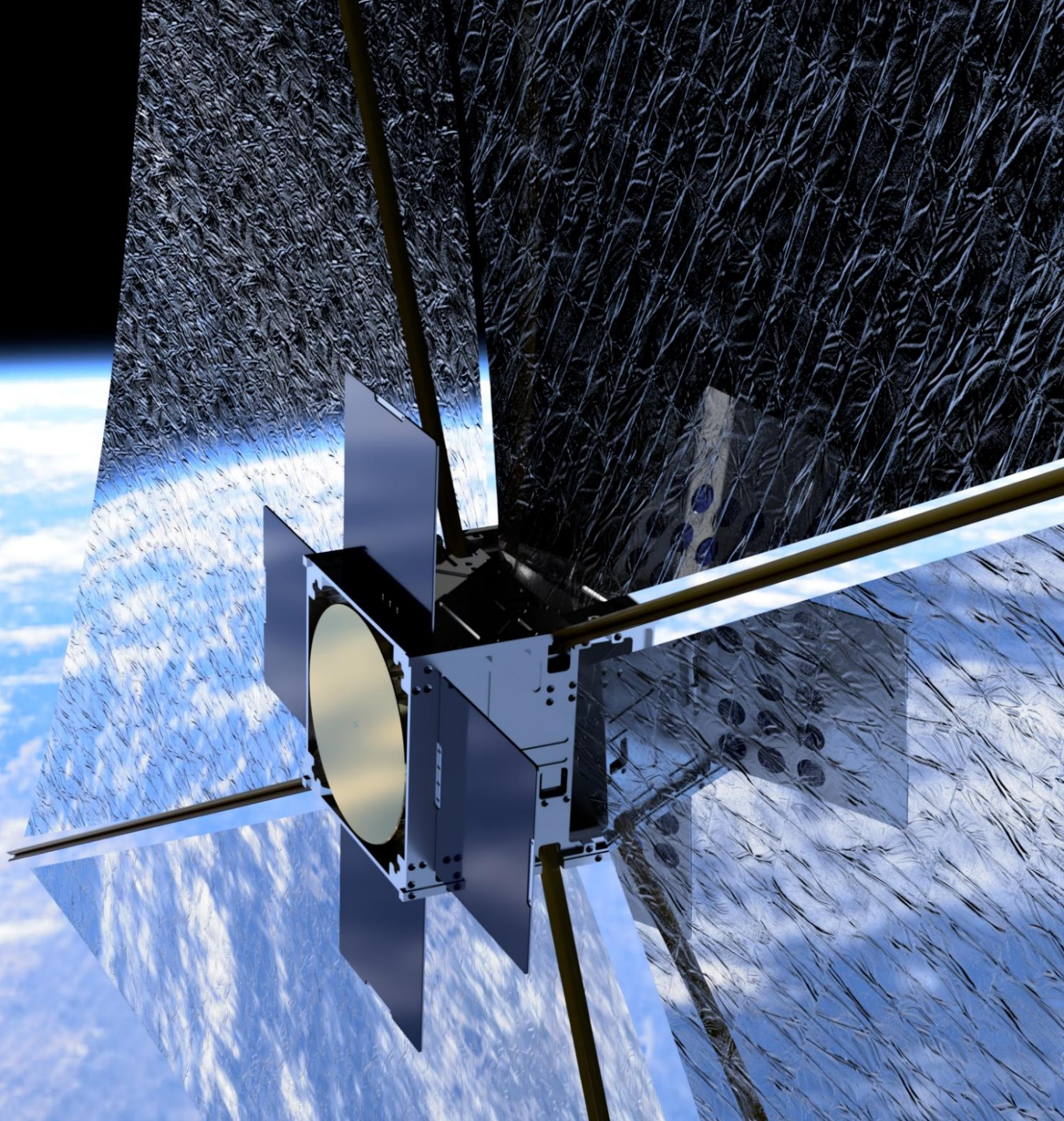





CAM1 str04 | 11:26-11:53
CAM4 str15 | 11:26-11:40
Deployment T-40s
CUBESAT str29+17 | 17:03-17:26

Our next steps:

Low-Earth Orbit Demonstrator

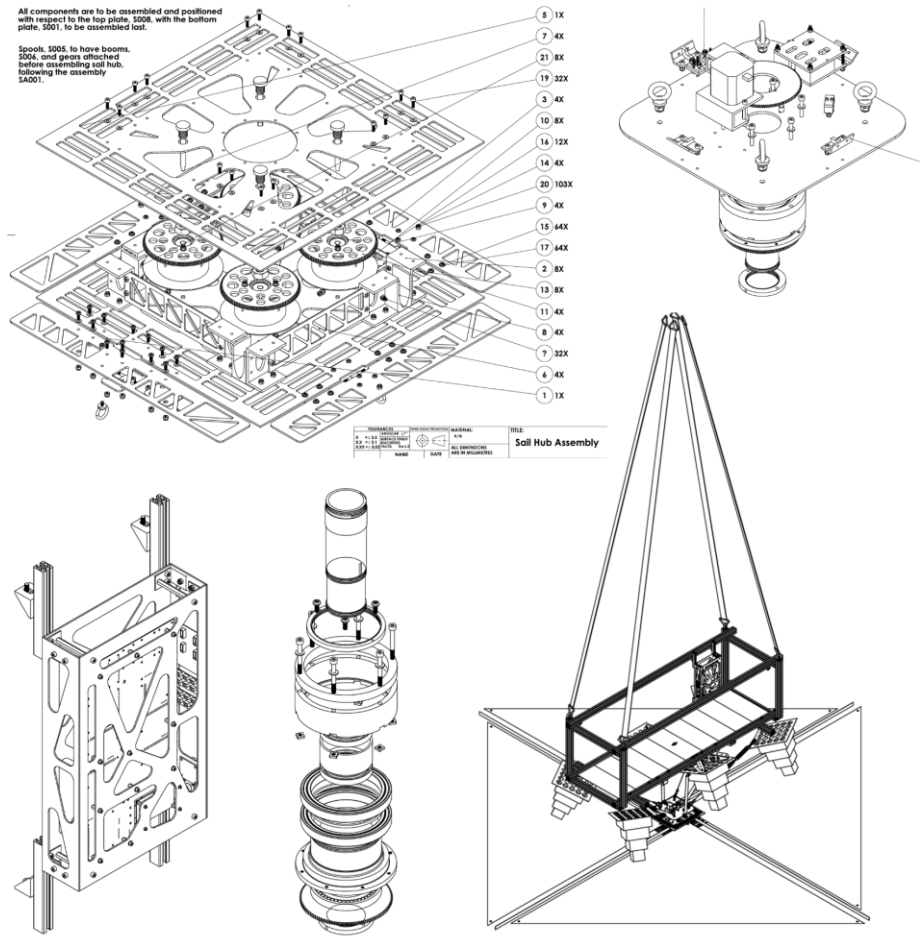


This year, we will work on development and assembly, integration, testing of a FlatSat

- ❑ Work on carbon composite boom hubs
- ❑ Develop space-grade electronics
- ❑ Test control systems for a satellite

For those new to engineering we will help develop you skills with guided satellite system projects

[redacted] PROJECT [redacted]



[redacted] SVAROG [redacted]

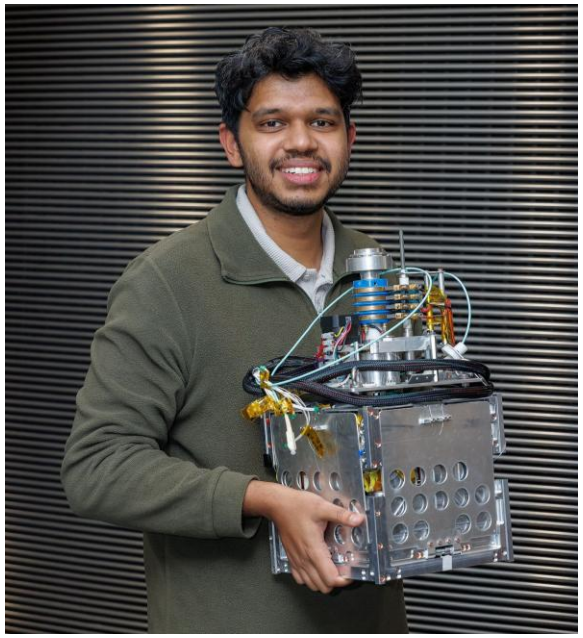
Meet the team | Executive Leads



Project Lead

Sahil Bhargava

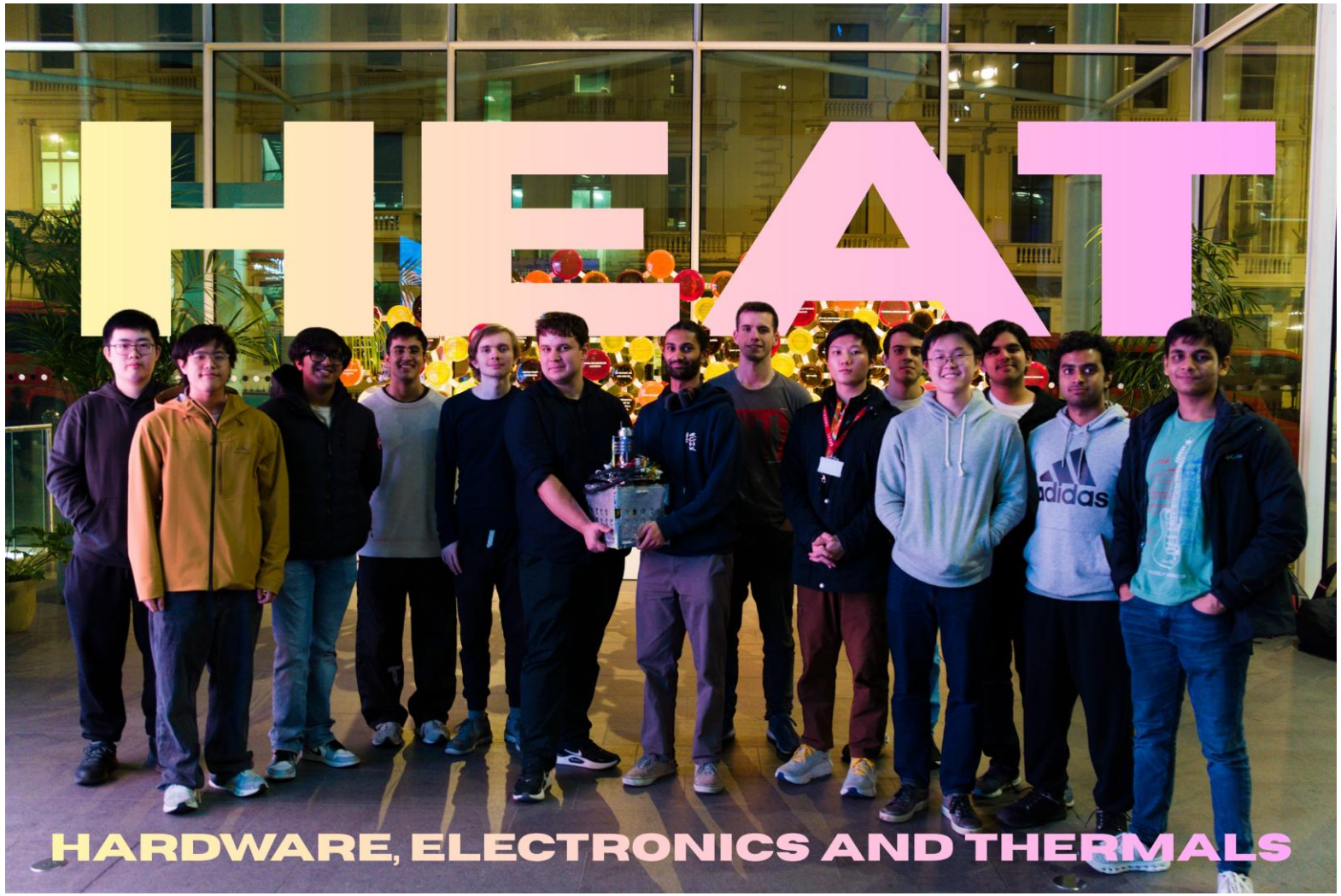
4th year Aeronautical Engineering Student



Technical Lead

Shiki Vaahan

5th year Aeronautical Engineering Student



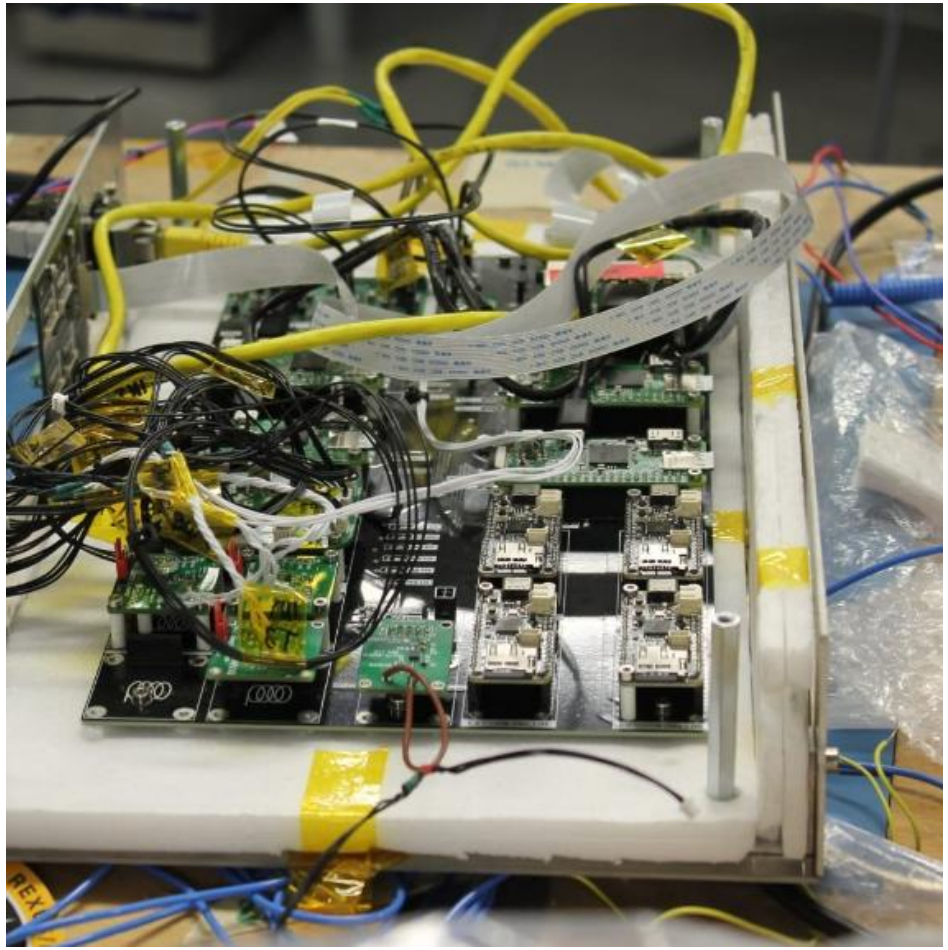


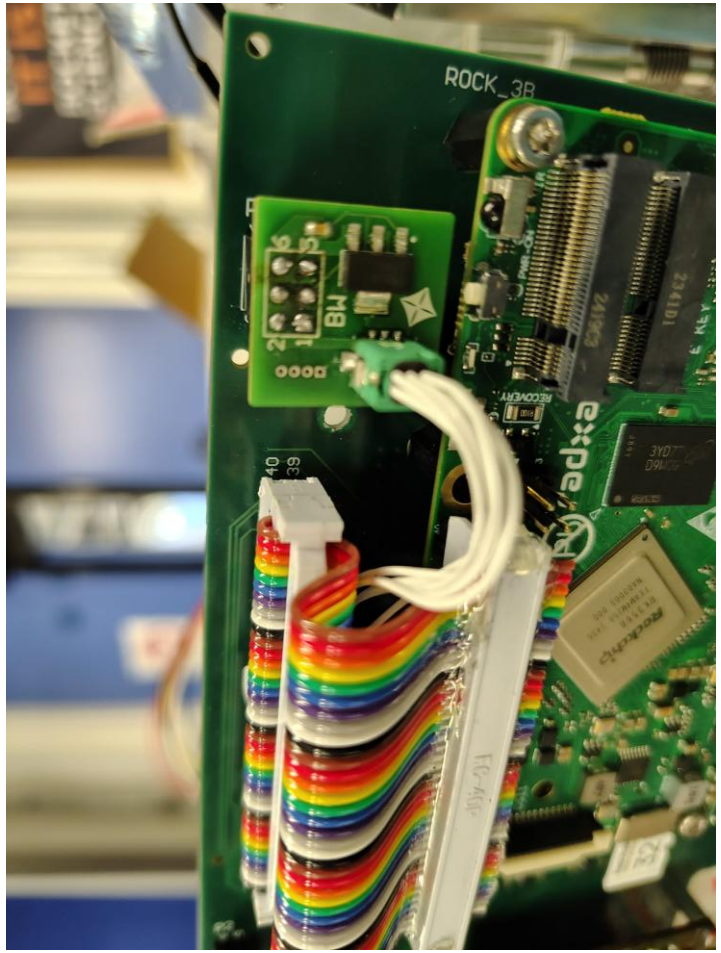
Electronics Lead

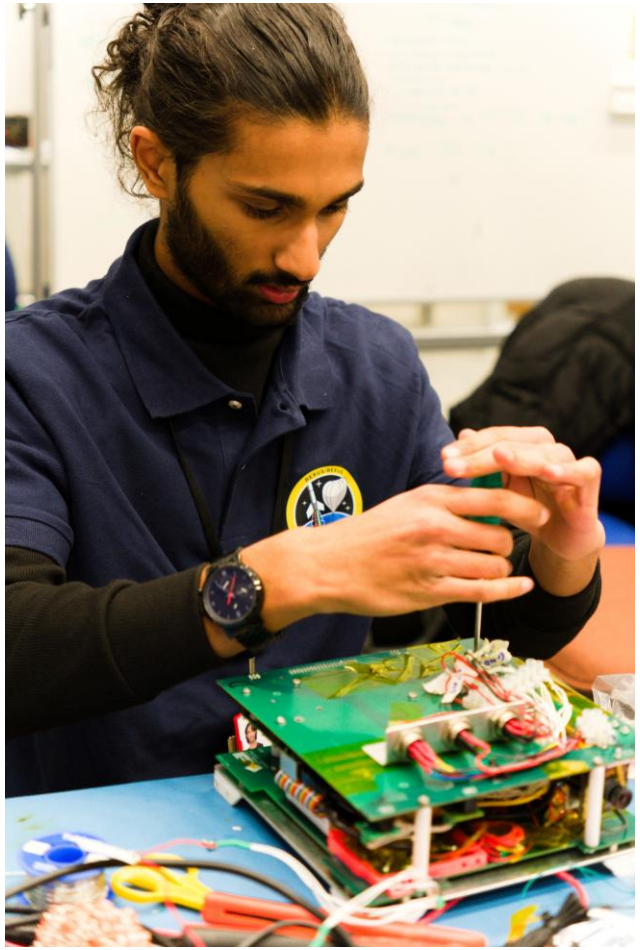
Nishant

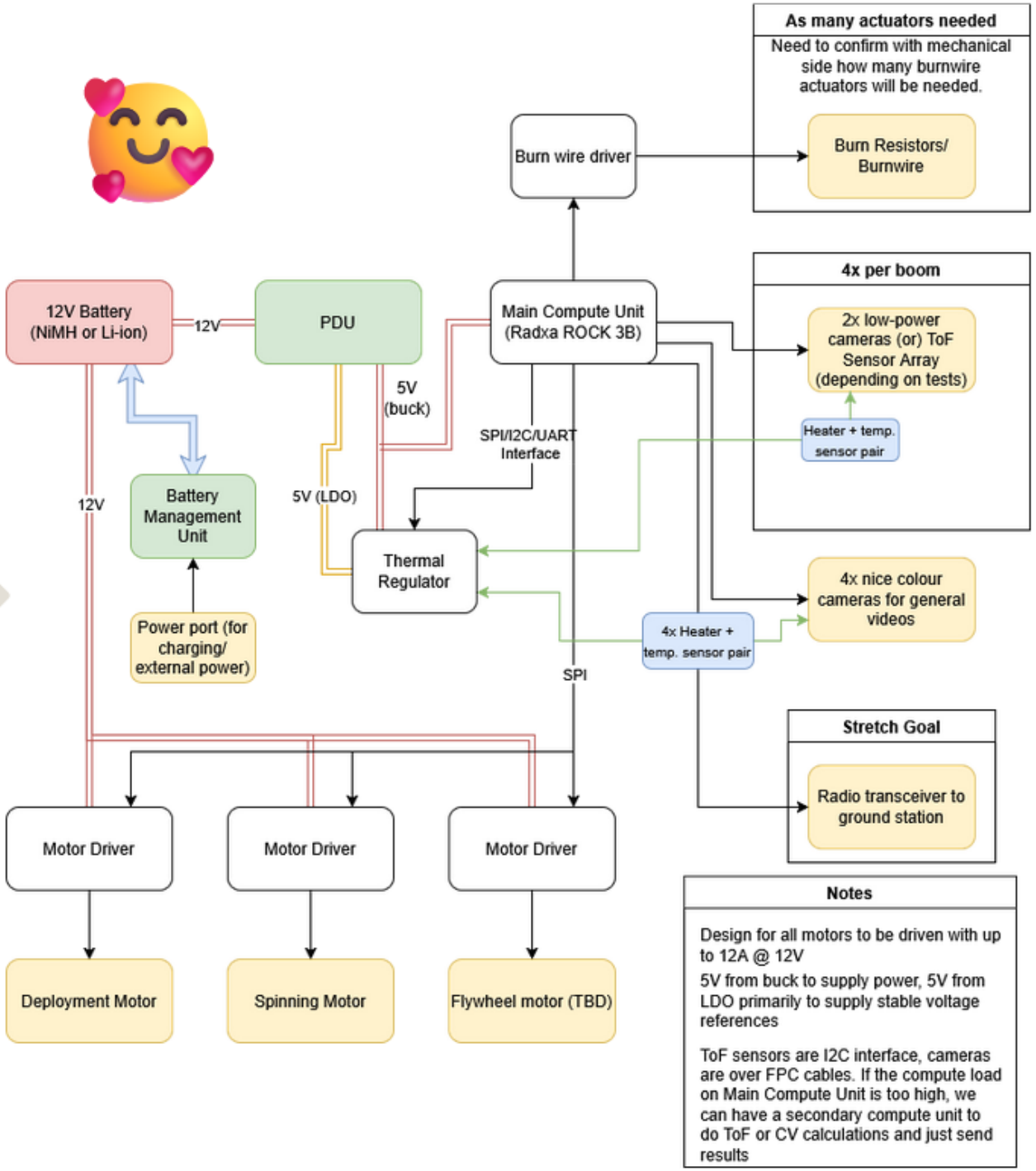
4th year Electrical Engineering Student

Designed and built the balloon-flight controller that commands boom deployment









Focusing on developing LEO ready electronics

- Battery Power System
- ADCS
- Communications
- Lower power design
- IR Boom Detection Array



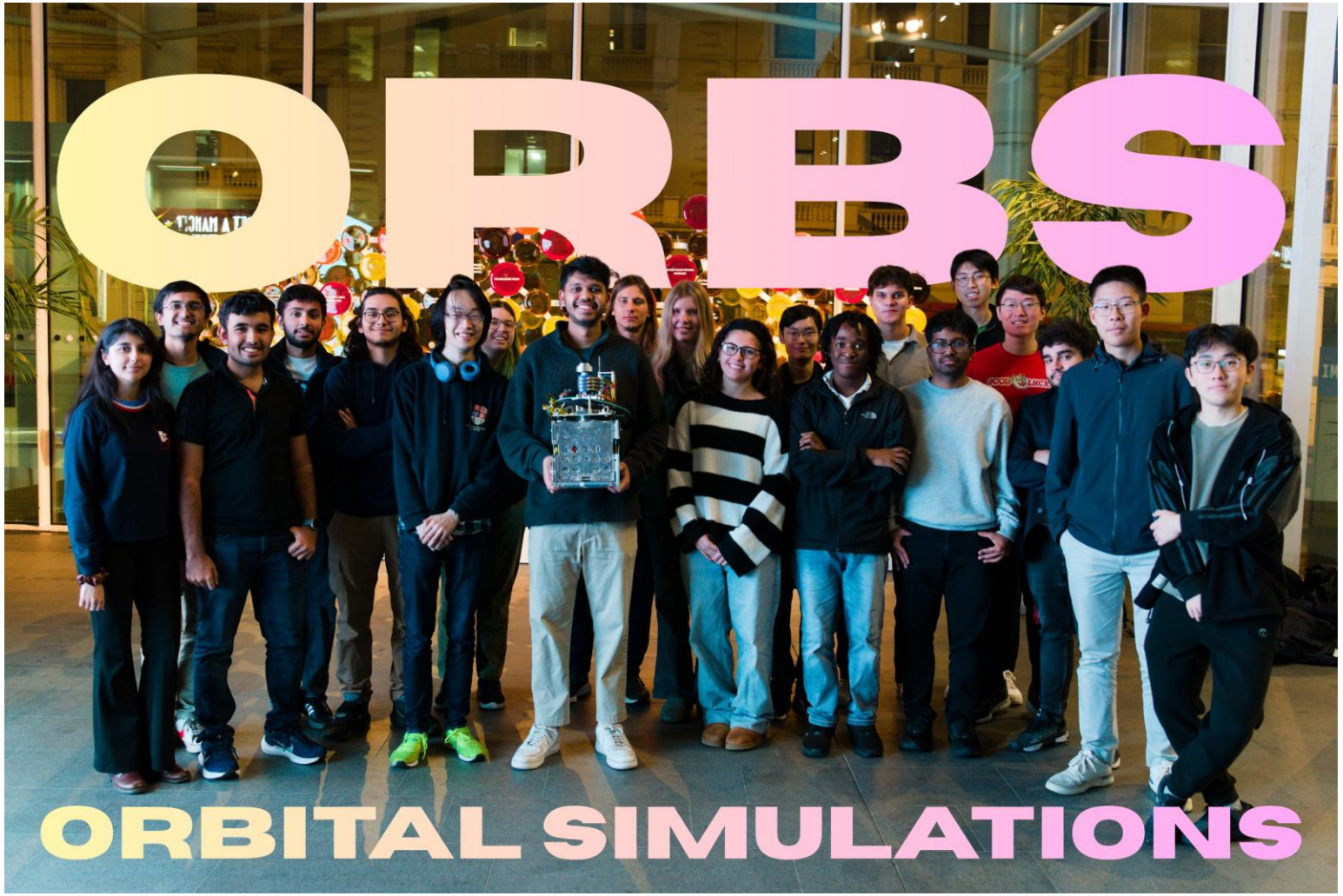


What we look for

- Regular commitment
- Work outside of meeting hours
- Basic electronics understanding (high school level)
- Curiosity and communication

What we give you

- Learn industry standard tools
- Resources and budget to experiment and prototype
- Fun stories to tell people
- Your hardware in space!





Orbital Lead

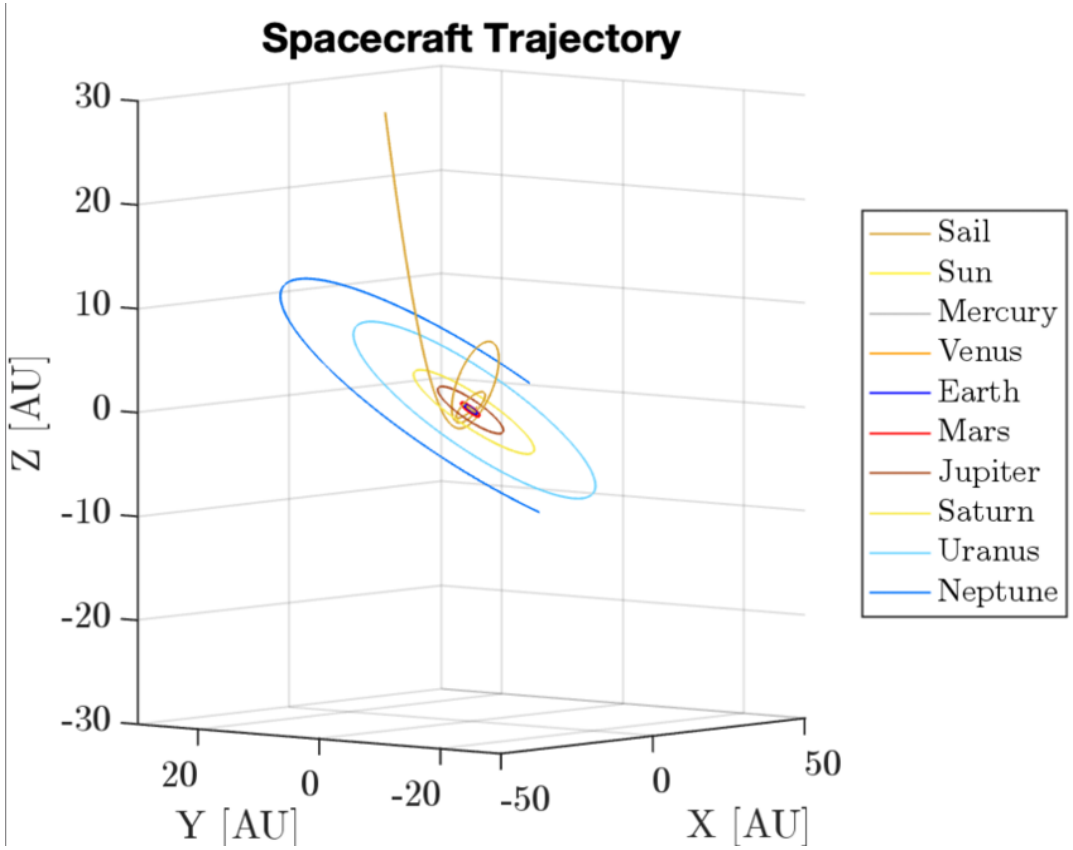
Rajveer Daga

3rd year Physics Student

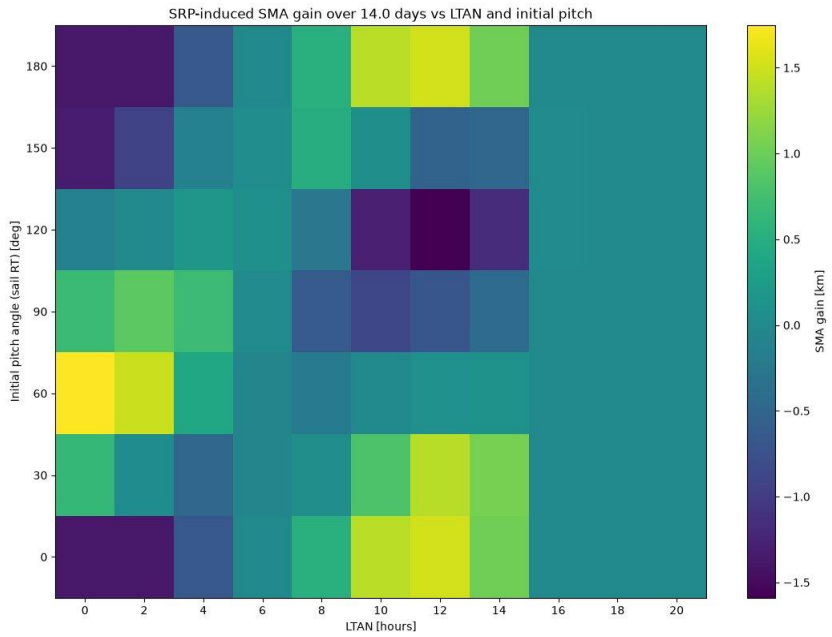
Developed in-house high-fidelity solar sail propagator models and published a conference paper at the International Solar Sailing Symposium

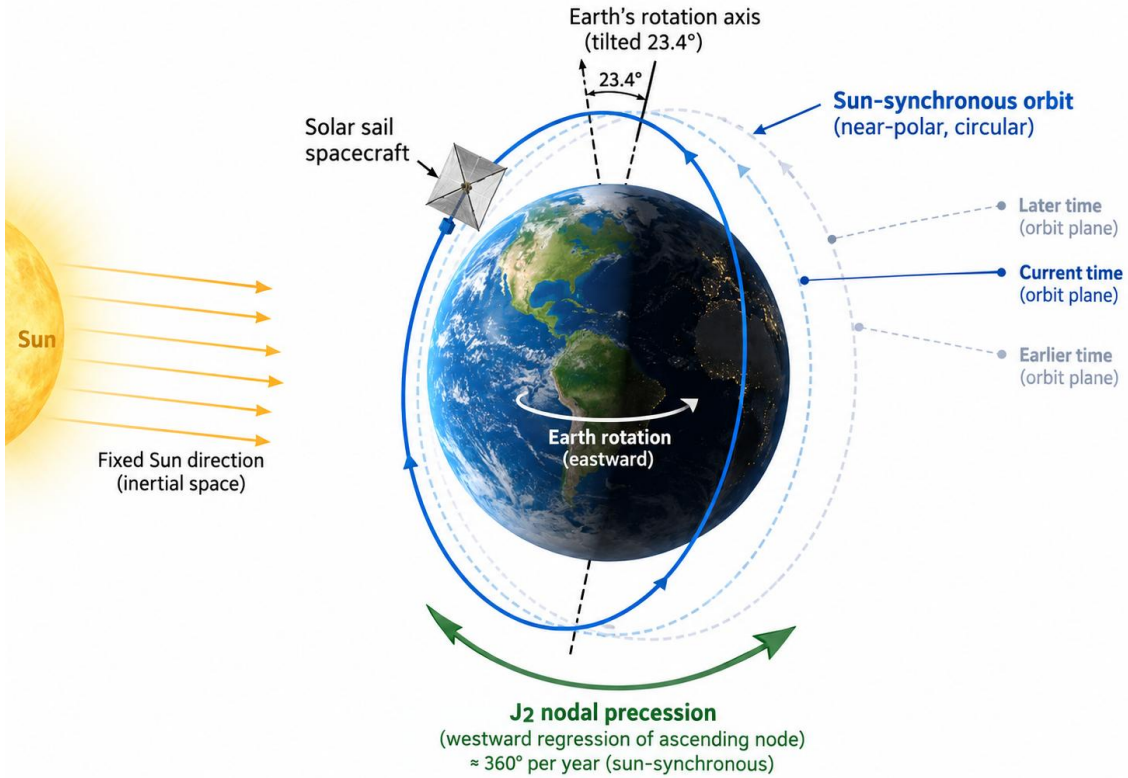
This year,

We will analyse solar system escape trajectories using our high fidelity propagator



Supporting Team with requirements for LEO demonstration





What we want

- Commitment (show up to meetings regularly)
- An interest in orbital mechanics and numerical methods
- Basic Coding Experience

What we offer

- You will gain experience developing and implementing numerical simulations
- Immersion into a research-like workflow
- Potential authorship on journal/conference papers
- An opportunity to learn Julia

7th International Symposium on Space Sailing (ISSS 2025)
Delft, The Netherlands, 30 June – 4 July 2025

Feasibility of Passive SMA Gain in LEO: Simulation and Stratospheric Demonstration of a Solar Sail CubeSat

Matthew Acevski[a,*], Shiki Vaahan[b], Piotr Fil[b], Herbie Christophers[c], Debdut Sengupta[b], Fatima Sajid[b], Madhav Rajeswar[a], Ian Poon[f], Rajveer Daga[a], Neil Pakrasi[d], Defne Ertugrul[e]

[a]Department of Physics, Imperial College London, London, UK
[b]Department of Aeronautics, Imperial College London, London, UK
[c]Department of Mechanical Engineering, Imperial College London, London, UK
[d]Department of Electrical Engineering, Imperial College London, London, UK
[e]Department of Chemical Engineering, Imperial College London, London, UK
[f]Department of Bioengineering, Imperial College London, London, UK

Abstract

Project Svarog is a student-led initiative based out of Imperial College London aiming to reach the heliopause using solar sail technology. The sail will be passively spin-stabilised and would not require gravity assists unlike previous interplanetary missions which makes deep space exploration significantly more feasible. The project is at a stage where we have developed and tested our first fully integrated mechanical prototype on the European Space Agency's BEXUS program (Balloon EXperiment for University Students), which allowed us to test deployment and rotation of the prototype in both near-vacuum conditions and extreme temperatures as well as validate our flight hardware. We also aimed to validate a scaling law (highlighted in [1]) for the behaviour of the sails under the influence of gravity. However, we were unable to complete this objective due to incomplete deployment of our sail on the BEXUS flight. Further, we have developed a high-fidelity comprehensive solar sail propagator model incorporating solar radiation pressure (SRP) and gravity-gradient torques - this model was developed to test the feasibility of trajectories for future launches into Low Earth Orbit (LEO) and interplanetary travel. In this paper, we discuss the developments and results of our orbital simulations as well as innovations in mechanical design made during the development of our BEXUS prototype in preparation for a future LEO or interplanetary mission.

Keywords: Deep Space, CubeSat, Solar Sail, Orbital Mechanics, Space Mechanisms

Nomenclature

BEXUS	Balloon Experiment for University Students
ESA	European Space Agency
SSC	Swedish Space Corporation
LEO	Low Earth Orbit
SRP	Solar Radiation Pressure
$\mathbf{a}_{\text{SRP}}$	Spacecraft acceleration due to SRP
$P_{1\,\text{AU}}$	SRP Flux at 1 AU
C_r	Solar Sail Reflectivity Coefficient
$\boldsymbol{\tau}_{\text{SRP}}$	Torque on Spacecraft due to SRP
$\mathbf{r}_{\text{lever}}$	Vector from Centre of Mass to Centre of Pressure
$\mathbf{F}_{\text{SRP}}$	Force on Spacecraft due to SRP
$\mathbf{a}_{\text{drag}}$	Spacecraft acceleration due to drag
C_d	Drag Coefficient
ρ	Atmospheric Density
$\boldsymbol{\tau}_{\text{gg}}$	Torque from Gravitational Gradient
μ	Earth's Gravitational Parameter
$\mathbf{R}(q)$	Quaternion-based rotation matrix
$\mathbf{I}_{\text{body}}$	Inertia Tensor of Spacecraft
ω_n	Spacecraft Nutation Frequency

*Corresponding author, matthew.acevski19@imperial.ac.uk

DEVELOPMENT, STRATOSPHERIC BALLOON FLIGHT TESTING, AND POST-FLIGHT FAILURE ANALYSIS OF A MOTORISED SAIL MEMBRANE DEPLOYMENT AND SPIN-TENSIONING SYSTEM

Gil Barbosa Ribeiro(1), **Herbie Christophers**(2), **Ivan Revenga Riesco**(2), **Dorota Budzyń**(3)
(1)*Lift Me Off, London, United Kingdom*
(2)*Imperial College London, London, United Kingdom*
(3)*ATG Europe/ESA, Noordwijk, Netherlands*

ABSTRACT

This paper presents the design and verification of an innovative deployment mechanism for a large-scale solar sail membrane, tested aboard the BEXUS 34 stratospheric ballooning mission. The system was engineered to evaluate the spin tensioning of a thin membrane while simultaneously validating novel deployable booms and their associated mechanisms under near-vacuum conditions. During the stratospheric flight, two of the four sail compartment doors failed to fully open due to the same failure mechanism, leading to incomplete deployment and preventing membrane deflection data collection. The study discusses the results of ground and flight tests, highlighting development and performance of the booms and anti-blooming mechanism, spin tensioning dynamics, and the electrical control of the mechanical actuators.

1. INTRODUCTION

Project Svarog [2,3] is a student-led initiative developing a cubesat for interstellar travel powered by a solar sail [1]. The stratospheric balloon flight with REXUS/BEXUS from Esrange Space Centre served as a first step technology demonstrator, allowing for a development and test cycle of a solar sail prototype.

REXUS/BEXUS allows university students to carry out their experiments either in suborbital rocket - offering two minutes of microgravity, or stratospheric balloons, with a minimum float of approximately one hour at an altitude of 25 km. The BEXUS platform was selected to take advantage of the challenging thermal and near-vacuum conditions as a stepping stone towards future in-orbit demonstrations. The BEXUS 34 flight experienced a float stage with the following environment: 16 mbar and -63 °C. The nominal deployed sail area would have been 5.8 m^2.

1.1 Noteworthy design constraints and requirements

The following were key constraints and requirements which drove the design:

1. Antenna cables for the gondola must pass through the centre of the assembly, requiring a 35 mm through hole;
2. Low temperature near-vacuum environment;
3. The gondola landing must be cushioned by crash pads. As the solar sail demonstrator is slung below the crash pads, the connection between the solar sail demonstrator and gondola should crumple upon touchdown;
4. Gravitational loads are still in effect at float altitude, and as such any deployed structures must be able to support their own weight (as opposed to orbital deployables).

2. MECHANICAL DESIGN

The mechanical design consists of two hubs connected via a shaft system. The upper one has the electrical connections and components, and is assumed fixed with respect to the gondola. On the other hand, the lower hub stows the sail and the booms, and has the capability of rotating about its axis.

2.1 Booms

A novel design of rollable boom was developed. 32 mm wide spring steel tapes were sourced from Stanley FatMax measuring tapes. Two of these concave tapes were then connected in a biconvex arrangement to increase the buckling and torsional stiffness. Initial attempts at connecting the tapes using adhesive Kapton tape (as an elastic hinge) revealed that the path length difference in the spooling of the inner and outer tape was significant enough over a 2 metre boom to buckle and tear the Kapton tape - Fig. 1.

To alleviate this, the tapes are connected by a non-adhesive Kapton sleeve with two axial pockets. This holds the tapes in the biconvex arrangement, yet allows them to slip axially relative to each other when spooled.

6th International Symposium on Space Sailing (ISSS 2023), New York, USA, 5-9 June 2023

Research for and Early-Stage Development of the First Interstellar CubeSat Powered by Solar Sailing Technology

Piotr Fil[a,*], Gil Barbosa Ribeiro[a], Debdut Sengupta[a], Beatriz Soriano Tortosa[b]

[a]Department of Aeronautics, Imperial College London, London, UK
[b]Department of Mechanical Engineering, Imperial College London, London, UK

Abstract

Project Svarog is a student-led initiative aiming to reach the heliopause using a solar sail [1]. The sail is set to be passively stabilised and does not require gravity assists unlike previous interplanetary missions, thus making deep space exploration more feasible and flexible. Previous feasibility studies have been performed, demonstrating the potential of the mission and highlighting research focus. A high-fidelity orbital model has been developed for proving the feasibility of the trajectory and studying initial conditions. Currently, Scientific Machine Learning [2] is being implemented to study the optimal initial conditions, parameters, and the sensitivity of the trajectory with respect to those properties of the system. Initial studies show that the escape trajectory is feasible for a mass to area ratio of 12 g m^{-2}. Given the repeated close passes to the Sun, the long duration of the mission, and its sensitivity to solar events, understanding and modelling the space environment for the duration of the mission is paramount. So far, preliminary simulations of radiation dose received by the spacecraft using GRAS [3] coupled with data driven model of solar activity have been performed. Structural simulations from an in-house code which uses multi-particle model have been compared with commercial packages and paired with vacuum chamber testing for validation. Following the IKAROS team research and analysis [4], we have now developed non-dimensional analysis which will enable scaling of sail dynamics to reduce number of required simulations and enable conducting experimental validation of sail behaviour under influence of gravity. Mechanical and electronic design and prototyping have been undergoing in parallel with the research endeavours. These have made testing of deployment methods and communications architectures possible. A motor-controlled boom deployment is being studied in parallel with the flight proven spinning method [5]. Should these technologies be successful, the Svarog system could serve as a low-cost enabler for the testing of new technologies and research opportunities in deep space, piggybacking of the increasing number of interplanetary missions and fostering deep space exploration.

Keywords: Deep Space, CubeSat, Solar Sail, Orbital Mechanics, Structural Design

Nomenclature

B	bit rate
c	speed of light
E	energy
$F, F*$	mass, characteristic scale
FEM	Finite Element Model
G	antenna gain
I	mean excitation potential
k, K	stiffness, characteristic scale
k_B	Boltzmann constant
l	length
L	Lagrangian
m, M	mass, characteristic scale
n	number of particles
N	noise
P	power
r	radius
t	time
T	characteristic timescale/temperature
x, y, z	position
X	characteristic lengthscale
z	charge of particle
α, δ	Euler angles
β	v/c
γ	Lorentz factor
λ	wavelength
Ω	spin rate

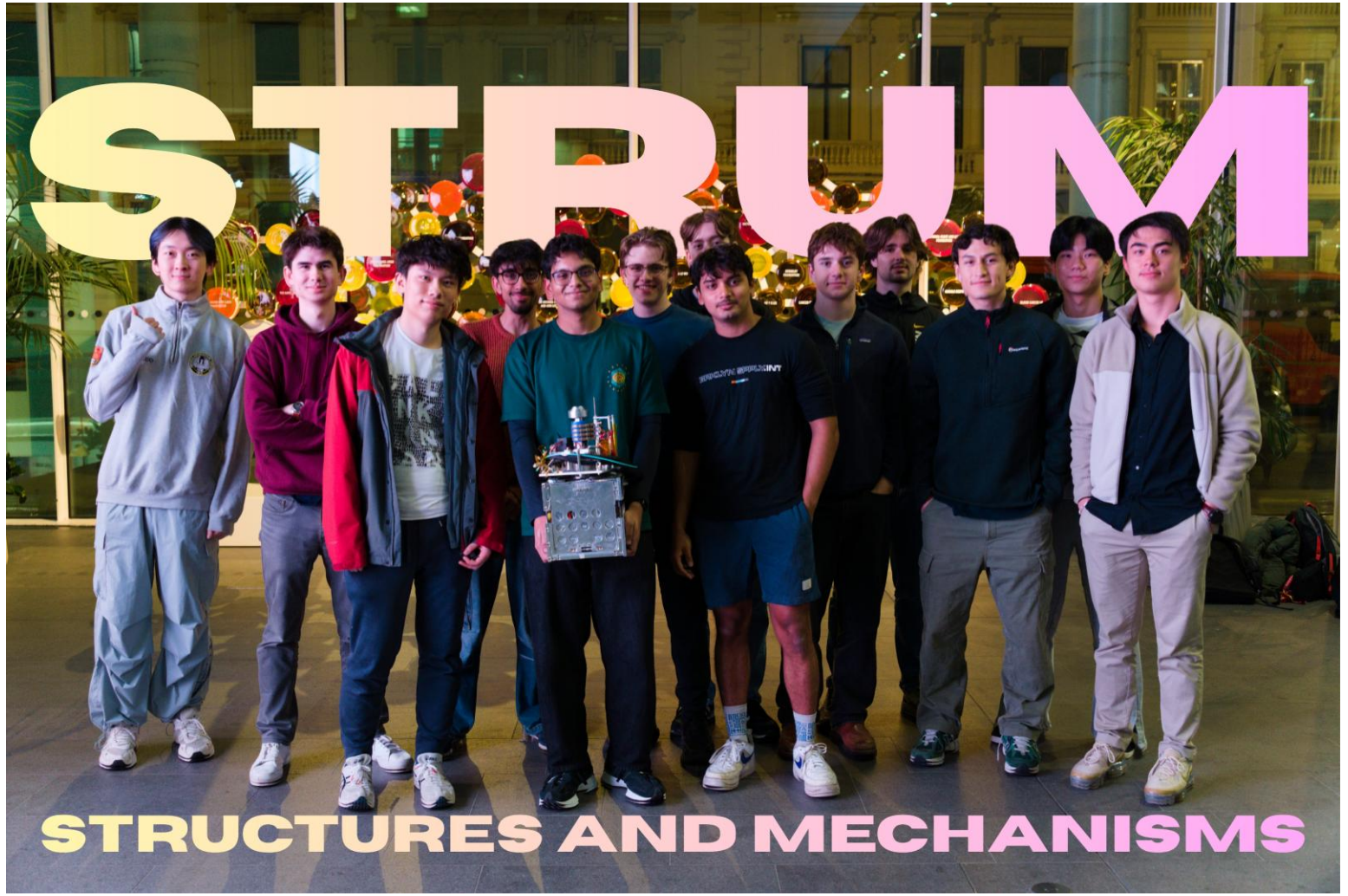
*Corresponding author, piotr.fil20@imperial.ac.uk



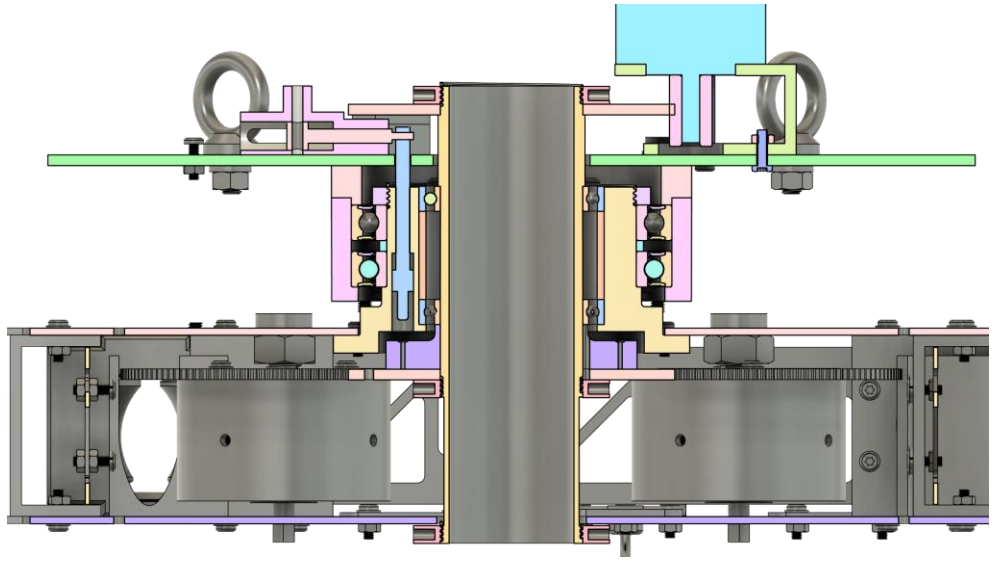
Structures Lead

Aditya Shankar

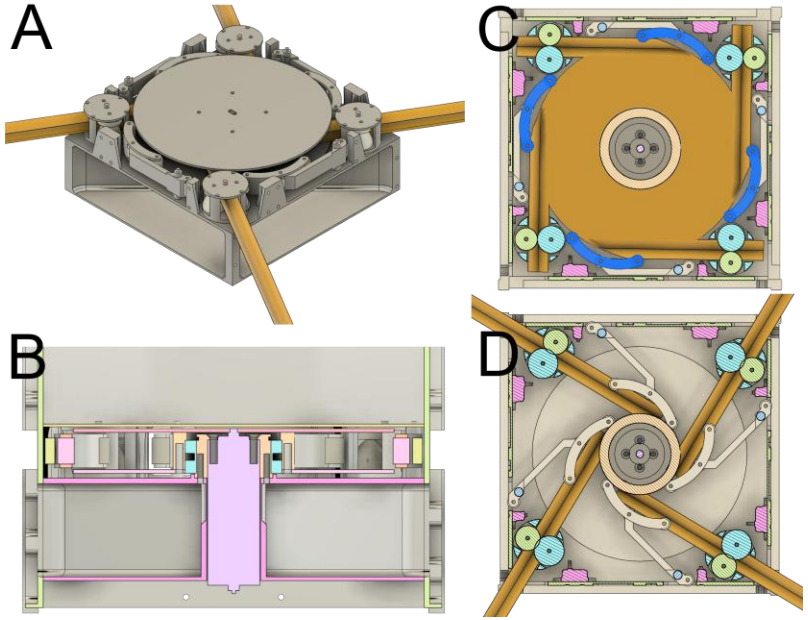
3rd year Aeronautical Engineering Student



Designed and built a solar-sail deployable structure entirely in-house



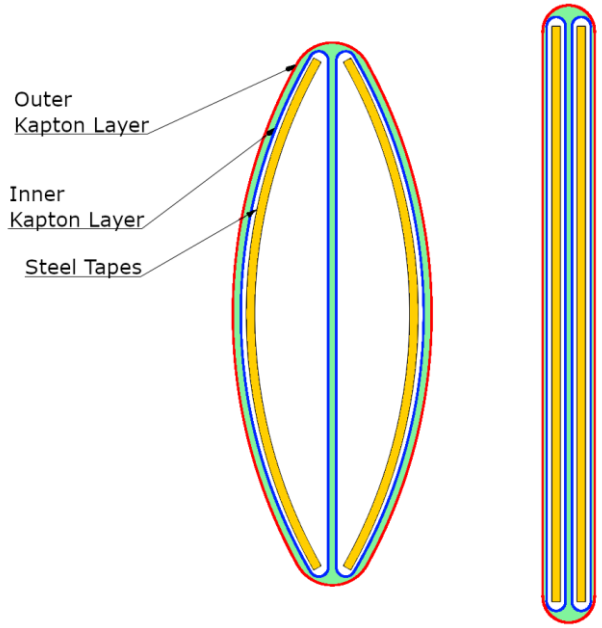






What have we done so far?

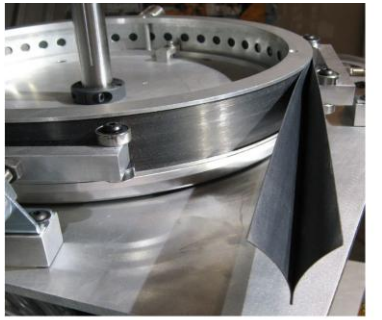
- ❑ **In-house boom design and fabrication and sail membrane design and stowage pattern developed**
- ❑ **Custom motor and gearbox deployment drive that worked in near-space conditions**
- ❑ **Achieved full deployment of solar sails on a European Space Agency balloon, among the few student teams with tested solar sail hardware**
- ❑ **One of the only student teams in the world to design composite booms**

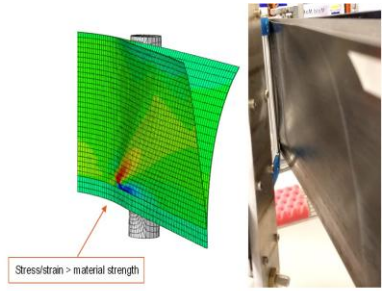


Goals for this year

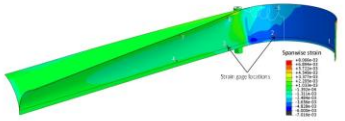
- ❑ Redesign of the frame for a 12U CubeSat
- ❑ Development of the flywheel mechanism
- ❑ Development and characterisation of the carbon composite booms
- ❑ Design and machine moulds for 1 m booms
- ❑ Aim for a Flat Sat deployment of the carbon composite booms







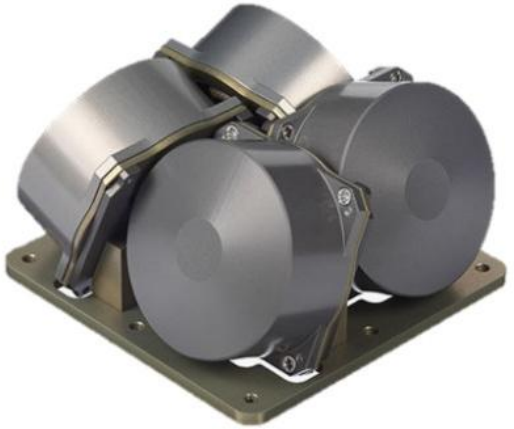


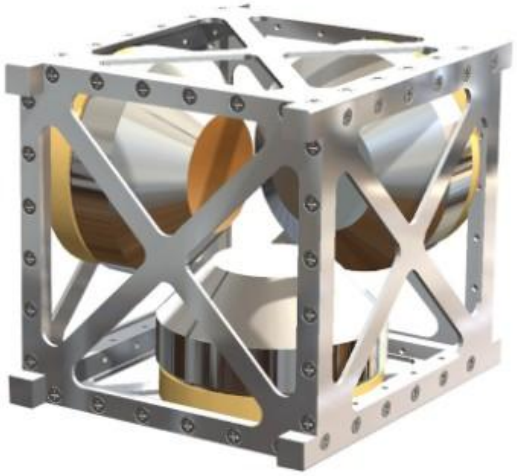



Now it’s time to
send our
hardware to orbit

Introductory Project

- ❑ Work on building a 2-axis flywheel 1U CubeSat Demonstrator!
- ❑ Split into teams of 10-15 people.
- ❑ Get to work on one of the following subprojects!
  - ❑ Simulations and Attitude Controls of the CubeSat
  - ❑ Electronics and Actuation Mechanisms
  - ❑ Structural Design and Assembly
- ❑ We will help and teach you what you need to know
- ❑ At the end of the project you will have the opportunity to build and test what you have worked on!





Introductory Project

- ❑ We expect that you will work in a fairly large group to complete the project. Groups should include members of both the mechanical, electronics, and orbital mechanics teams.
- ❑ No prior experience is required and we will teach you the basic knowledge required to start the project. The project leads are always available to help and give guidance if you are stuck. As leads we want to foster a high level of skill within the teams, so we are more than eager to help if needed. We do, however, expect a decent level of commitment to the project (whatever is required to meet the timeline below).
- ❑ Use GitHub for any code, and project management on Notion/Monday.com. Code on Python/C++. Simulations on MATLAB Simulink / Python. CAD on Fusion 360.
- ❑ At 3 weeks of the intro project running, we will ask for the group to decide on an electronics, mechanical, and software lead. These will be the people responsible for communicating to each of the other members in the respective departments and will be the primary contact point for the leads (obviously we will still be talking to each of you, but we expect leads to be the ones generally knowledgeable).
- ❑ At the end of the project, we will be 3d printing all your designs out, to do testing of all the designs! Make sure all your designs are made with DfAM in mind.
- ❑ Testing and scoring will be based on a mix of CAD and Coding etiquette and efficiency, RMSE of control system along with total Energy draw. We will share a proper Rubric in the near future!

Introductory project timeline

Date	Event	Description
12-10-26	1st meeting	Form the group, understand tasks and group member's experience and expertise. Q&A with leads
18-10-26	2nd meeting	Draw out project timeline, milestones, etc. Expect project members to be building any prerequisite.
1-11-26	Subsystem Leads proposal	Group will chose 3 representatives for electronics, mechanics, software
30-11-26	PDR	Preliminary design review. We expect individual subsystems (eg: frame, motors, software) to be in good shape (80% done), but low/no integration work started.
11-01-27	Post-winter break status update	Subsystem leads (primarily) present progress since PDR (we assume there will be less/none) and roadmap for next weeks (important!)
01-02-27	Project conclusion	Work is finished on the project. This weekly meeting will be devoted to showcasing and testing the project out.

Why Join Us?

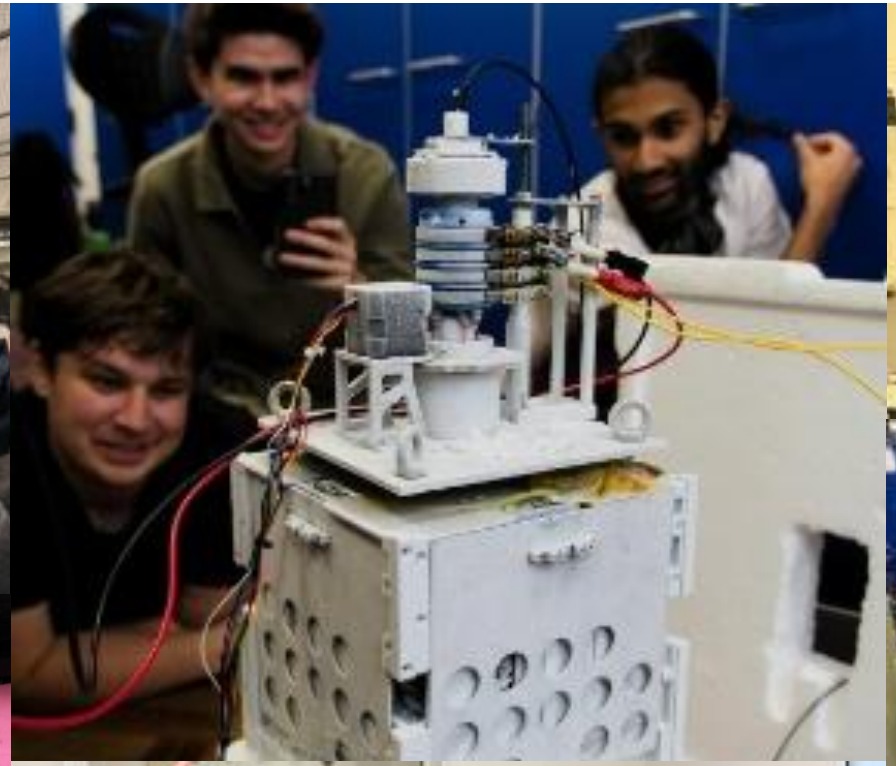
- ❑ Tons of hands-on experience
- ❑ Learn about manufacturing, design, assembly, integration and other engineering workflows.
- ❑ Gain an insight into writing and producing academic texts and get to contribute to publishing papers
- ❑ Trips to ESA for testing and trips to conferences to present papers
- ❑ Networking opportunities during reviews, launches and events
- ❑ **Most importantly, get to see something you build reach LEO**



This is the European Space Tech Landscape as of this September

If that hasn’t convinced you here are some other highlights

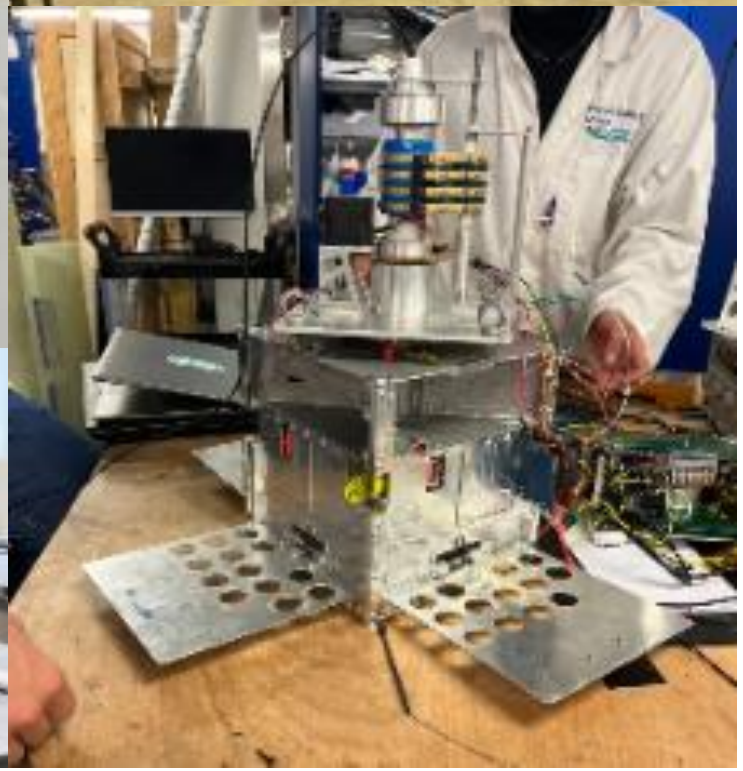


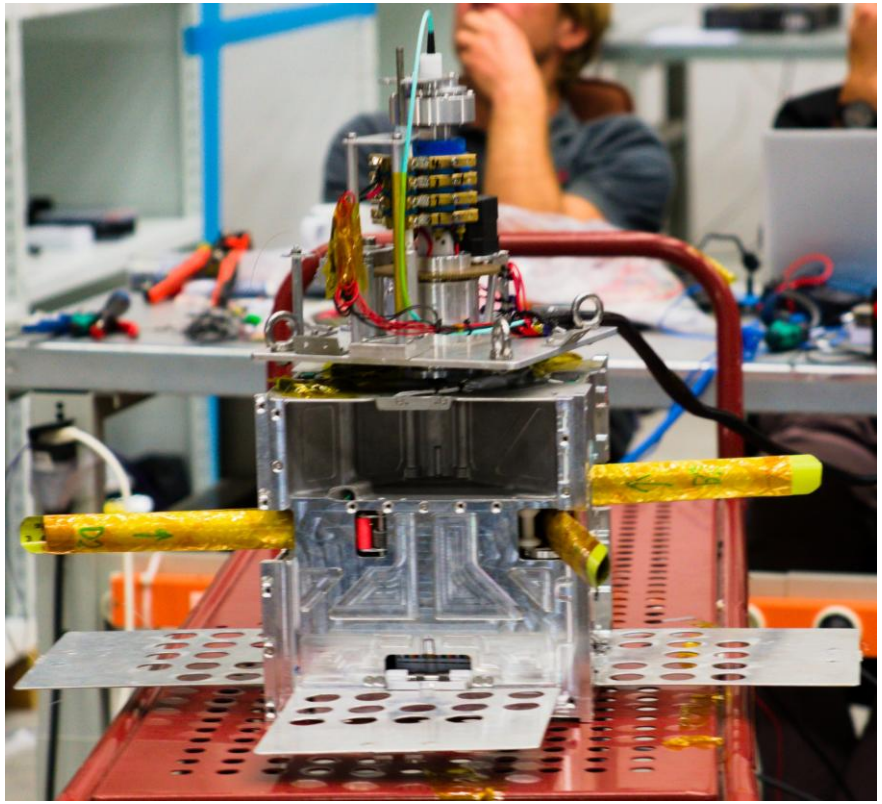












It is deployment time!

1. **Burn wires**
2. **Boom deployment**


PROJECT
SVAROG
BEXUS 36 CYCLE 16
PROJECT SVAROG²

LinkedIn

PROJECT SVAROG

Instagram



Subteam

indication form

1. Electronics
2. Structures
3. Orbital/Controls
4. Introductory project